INCH-POUND
MIL-DTL-24643/9F
1 October 2009
SUPERSEDING
MIL-DTL-24643/9E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, TYPE LSMRI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, coated or uncoated, Class K stranding (see table I for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Minimum average wall thickness of 0.020 inch. Standard identification code applied by Method 3.
Fourth	-	The specified number of conductors shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/9	Type and size	Conductors		Insulated conductor diameter		Conductor resistance (ohms)
		No.	AWG	min. (inch)	max. (inch)	
-01UN	LSMRI-D-1	2	20	0.068	0.075	11.05
-02UN	LSMRI-D-2½	2	16	0.092	0.102	4.36
-03UN	LSMRI-T-2½	3	16	0.092	0.102	4.36

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor, 20 AWG	1000
Conductor to conductor, 16 AWG	1500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor (all sizes)	100
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

Group C:

Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Heat distortion (percent of unaged, max.)	30

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.

Group D:

Flame propagation (cable)	Not applicable

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Acid gas equivalent (percent, max.)	
Insulation	18
Halogen content (percent, max.)	
Insulation	0.2
Smoke index (max.)	
Insulation	45
Toxicity index (max.)	
Insulation	1.5
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMRI-D-1	600
LSMRI-D-2½	600
LSMRI-T-2½	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-013)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.